

Search Frictions and Welfare in a Market for Prominence: Theory and Experiment*

Zitian Wang[†]

September 30, 2026

[Click here for the latest draft](#)

[Click here for the latest slides](#)

Abstract

In online marketplaces, sellers pay platforms to make their products more visible and easier to inspect, creating a market for search prominence. I study how search friction affects the level and distribution of surplus in this market. In my model, two *ex ante* identical sellers bid for a prominent position in a second-price auction and then simultaneously post prices. The consumer observes both prices and learns her match value for the prominent seller's product for free before deciding whether to incur a search cost to inspect the second seller's product. I derive equilibrium bids, price distributions, and search cutoffs, and test these predictions in a laboratory experiment with low and high search costs. Platform auction revenue changes little as search cost rises, despite the predicted increase. Total and consumer surplus decrease with search cost, but consumer surplus falls proportionally much less than predicted. At both search costs, consumer surplus exceeds equilibrium benchmarks, whereas sellers jointly receive only about a fifth of total surplus after paying for prominence. Further analysis of bidding, pricing, and search behavior helps explain these findings.

JEL Codes: C72, C91, D83

Keywords: consumer search, paid prominence, price competition, welfare, experiment

*I am grateful to my advisors Charles Holt, Po-Hsuan Lin, and Peter Troyan for their guidance.

[†]University of Virginia, Department of Economics. Email: zw3qt@virginia.edu.

1 Introduction

Online marketplaces offer a wide range of products, but these products are not displayed equally prominently. Some appear at the top of search results or receive more visible display space, while others require consumers to scroll through several pages of results. Sellers pay platforms to make their products easier to discover and inspect, creating a market for search prominence. Google search ads, Amazon Sponsored Products, and sponsored hotel listings are familiar examples. These arrangements give sellers a way to compete for consumers' attention in addition to competing for their purchases. A seller pays for an opportunity to be considered, while the return to that payment depends on what consumers do after encountering its product.

Costly consumer search gives sellers a reason to compete for prominence. Consumers spend time and effort learning whether a product suits their needs and may stop before inspecting every alternative. A prominent seller can therefore reach buyers who never inspect its competitors. The value of this advantage depends on how costly those competing products are to inspect. If examining an alternative requires substantial effort, a consumer may purchase the product already inspected even though another product could offer a better match. Prominence consequently has value even when it conveys no information about product quality. It changes the cost of reaching a seller relative to its competitors, and thereby changes the demand that seller can expect to receive.

Readily available price information also allows sellers to influence consumers' search decisions. Prices appear in product listings and price advertisements, so consumers can often compare them without first undertaking a detailed inspection. Learning whether a product is a good match can nevertheless remain costly. By posting a price, a seller changes both the surplus from purchasing its product and the consumer's incentive to inspect another product. A low price at a competing seller may make further inspection worthwhile, whereas an attractive price at an already inspected seller may encourage the consumer to stop. Prices thus help determine the products that enter consumers' consideration sets, as well as their choices within those sets. The value of prominence depends on this pricing response to the search environment.

This paper asks how search friction affects the level and distribution of surplus when sellers compete for prominence. I develop a model linking bids for a prominent position to subsequent pricing and consumer search, and test its predictions in a laboratory experiment. The experiment measures platform revenue, consumer surplus, and seller profits within the same market. This makes it possible to distinguish changes in the surplus created through trade and search from transfers between consumers, sellers, and the platform. The distinction matters because a valuable position need not leave its winner with a large profit after payment. Similarly, an increase in the cost of inspecting competing products can affect the platform through the value of the position it sells, in addition to affecting consumers through their

search expenditure.

A laboratory setting is useful because the meaning of prominence is difficult to isolate in the field. Rankings can reflect expected relevance or product quality, and personalized recommendations can convey information about a consumer’s likely match. A prominent product may then attract inspection both because it is easier to find and because consumers expect it to be more suitable. In the experiment, sellers have identical distributions of match values and do not know their realized matches when they bid or price. Prominence provides a structural search-cost advantage: the first product is inspected for free, whereas inspecting the second costs the buyer a known amount. I vary that cost while holding the remaining market rules fixed. This isolates the inspection-cost advantage without changing the information conveyed by the position about product fit.

The model has two *ex ante* identical sellers and a buyer with unit demand. Sellers first compete in a second-price auction for a prominent position and then simultaneously post prices. Each product is independently a good or bad match for the buyer. The buyer observes both prices and learns the prominent product’s match value for free. She then decides whether to pay to inspect the second product, after which she may buy from either inspected seller or exit. Prominence gives one seller the initial inspection, but does not restrict the buyer’s choice among products she has already inspected. Sellers anticipate the resulting demand when they price, and anticipate the revenue difference between the two positions when bidding for prominence.

The buyer’s optimal search rule depends on the initial match. After a good match, the buyer already has a product she values. Further search is worthwhile only if the possible saving from the second seller’s lower price compensates for the inspection cost. The relevant cutoff is therefore in the difference between the two prices. After a bad match, the buyer instead searches for a product worth buying, so the decision depends on the second price alone. These two branches of demand give the sellers different opportunities to compete. The prominent seller can use its price to retain a buyer who has found a good match, while the second seller’s price affects both its ability to attract that buyer and its ability to reach a buyer disappointed by the first product.

These search rules generate opposing pricing incentives. On a segment where the buyer’s search decision does not change, a seller benefits from raising its price. A price reduction that crosses the search cutoff can instead produce a discrete increase in demand. I show that the pricing subgame has no pure-strategy Nash equilibrium and construct an equilibrium in mixed strategies. Under the experimental parameters, the prominent seller’s price distribution first-order stochastically dominates the second seller’s distribution. The prominent seller charges more on average because consumers can inspect its product without paying the cost required to reach its competitor. Both the price distributions and the demand generated by the buyer’s stopping decisions enter the expected revenue advantage of prominence.

In the constructed equilibrium, this revenue advantage equals the search cost. It is also the seller's willingness to pay for the top position rather than remain second. The second-price auction therefore has a symmetric equilibrium in which both sellers bid the search cost, and platform revenue increases one for one with that cost. Although the top seller earns more revenue in the product market, its position payment eliminates this expected advantage in net profits. The model thus provides linked predictions for bids, prices, search, and the division of surplus. Observing only auction payments would leave unanswered how the underlying position values were generated, while observing only product-market outcomes would omit the transfer from the winning seller to the platform.

The experiment implements the game with low and high search costs of 20 and 40 points. A good match is worth 100 points and occurs independently with probability one half for each product. Across six sessions, 87 subjects make bidding, pricing, search, and purchase decisions over 30 rounds, yielding 870 three-player markets. Roles remain fixed, while sellers and buyers are randomly rematched each round. Match values are redrawn, and subjects receive feedback on realized market outcomes. The design observes sellers' willingness to pay for the position as well as the subsequent decisions that determine its return. The auction provides a simple institution for allocating prominence and collecting payment, rather than reproducing every feature of a particular platform's advertising system.

I organize the results around welfare and then examine the behavior behind the welfare findings. For each experimental market, I calculate the equilibrium outcome conditional on its realized match values, taking expectations over the equilibrium price distributions and the implied search and purchase decisions. This comparison accounts for differences in the frequency of good and bad matches across treatments. The observed and predicted outcomes therefore face the same realized opportunities for trade. Total surplus records the value of the product purchased less the cost of any additional inspection. Its decomposition into consumer surplus, seller profits, and platform revenue distinguishes departures in the creation of surplus from departures in its distribution.

The first finding concerns platform revenue. Auction revenue changes little between the two search costs, despite the predicted increase of 20 points. The losing bid, which determines the winning seller's position payment, averages 23.35 points at the low cost and 23.83 points at the high cost. Average submitted bids do increase, but by only about 4.8 points. Thus, the weak response of auction revenue accompanies a much smaller response of sellers' elicited willingness to pay than the model predicts. The distinction between submitted bids and payments is useful here. The former measure sellers' offers for prominence, whereas the latter are the actual transfer to the platform and the amount deducted from sellers' product-market revenue.

The second finding is that consumers fare better than the equilibrium benchmark suggests. Consumer and total surplus are lower at the high search cost, but consumer surplus exceeds

its benchmark at both costs. It falls by 27.2 percent from the low-cost treatment to the high-cost treatment, compared with a predicted decline of 78.7 percent. These are distinct comparisons: consumers receive more than predicted within each treatment, and their surplus responds proportionally less to the increase in search cost. Neither conclusion requires total surplus to exceed equilibrium. In fact, total surplus falls below its benchmark at both costs. Consumers can receive a larger share of a smaller surplus when their payments to sellers are sufficiently lower. The welfare decomposition makes this distinction visible rather than treating higher consumer surplus as evidence of a more efficient allocation.

The third finding is that sellers jointly retain around 20 percent of total surplus after paying for prominence at either search cost. Their combined profits fall well below equilibrium at the low cost and are close to equilibrium at the high cost. Combined sales revenue, however, falls below the equilibrium benchmark in both treatments. At the low cost, the revenue shortfall accounts for most of the profit shortfall, with payments above the predicted level adding to the loss. At the high cost, payments far below the predicted 40 points largely offset the revenue shortfall. The similar observed profit shares therefore arise from different comparisons with equilibrium. A small net profit gap at the high cost does not mean that either product-market revenue or position payments are close to their predictions.

Pricing and search behavior help explain these distributional outcomes. Prominent sellers charge more on average than second sellers, but their prices are substantially lower than predicted at both costs. Prices at both positions are also more dispersed than predicted. Buyers with a good initial match largely follow the theoretical search cutoff. Lower prominent prices can therefore benefit these buyers twice: they reduce the payment when the buyer purchases from the top seller and can encourage her to stop without incurring another search cost. After a bad initial match, some second prices exceed the cutoff at which further inspection is worthwhile, leading to exit even when the uninspected product would have been a good match. Compliance with the search rule is also lower in this branch, especially at the high cost, with both under- and over-search. These findings connect the welfare comparisons to observed decisions without attributing all deviations to a single behavior.

The contribution is to experimentally elicit willingness to pay for prominence in a market where pricing and consumer search determine the value of the position. Jointly observing these decisions allows bids to be compared with both the equilibrium position value and the revenue advantage delivered in the experimental markets. It also shows how departures from the pricing and search predictions affect the distribution of surplus after sellers pay for their positions. The treatment varies consumer search cost, not a value assigned directly to the auctioned object. This distinction connects competition for prominence to the underlying friction that makes an inspection advantage valuable and allows the experiment to test the model's connection between the auction and the product market.

1.1 Related literature

The paper builds on models in which search frictions affect price competition. Diamond (1971) shows how costly price information can sustain monopoly pricing. With differentiated products, search also allows consumers to find a better match, as in Wolinsky (1986) and Anderson and Renault (1999). When prices are observable before inspection, they can influence which products consumers inspect and whether search continues. Choi et al. (2018), Ding and Zhang (2018), and Obradovits and Plaickner (2023) study pricing under this information structure. The present model retains costly learning about product fit with publicly observed prices, but sellers acquire the initial inspection through an auction. A low price by the second seller can induce continued search, but does not move that seller ahead of the inspection already supplied by prominence. This separates the acquisition of the top position from the subsequent use of prices to influence continued search, while allowing both decisions to respond to the same search friction.

Prominence and ordered-search models explain how an advantage in reaching consumers changes price competition. Armstrong et al. (2009), Zhou (2011), and Armstrong (2017) study pricing when consumers encounter firms in a particular order. Armstrong and Zhou (2011) examine how firms obtain prominence, including through payments to intermediaries and competition in advertised prices. These papers connect the value of being inspected early to subsequent pricing and consumers' continuation decisions. Field evidence also supports distinguishing inspection from purchase. Using hotel-ranking variation, Ursu (2018) provides evidence for a search-cost channel through which rankings affect inspection. My focus is how the cost of inspecting competing products affects the value of a purchased inspection advantage and the distribution of surplus. The experiment makes the difference in inspection costs explicit and holds sellers' match distributions fixed, so a winning position does not signal better product quality or a better match for the buyer.

Competition for prominence also connects the paper to sponsored-search auctions. Edelman et al. (2007) and Varian (2007) analyze auctions allocating advertising positions with different click volumes. Chen and He (2011) and Athey and Ellison (2011) incorporate consumer search and the information conveyed by paid rankings. The closest theoretical comparisons for endogenous position values are Xu et al. (2011) and Anderson and Renault (2026). Xu et al. (2011) derive position values from subsequent price competition, while Anderson and Renault (2026) show how firms' incentives to move in the search order depend on competition from firms ahead of and behind them. I bring this connection into the laboratory by eliciting bids and observing the pricing and search decisions that generate the position's return. The second-price auction is used to implement competition for the inspection advantage. The central experimental comparison concerns how the value of that advantage responds to search cost, rather than how bidding differs across auction formats.

Experimental studies of search and pricing provide the basis for examining the behavior

that generates these position values. Davis and Holt (1996), Abrams et al. (2000), and Cason and Friedman (2003) test how search and information frictions affect prices and market performance. Brown et al. (2018) examine sellers' use of exploding offers to discourage further search when prices are visible. Their experiment shows why consumers' behavioral responses matter for the profitability of a strategy intended to prevent continued search. When prominence is purchased, pricing and search behavior also determine the value of the object sellers bid for. The present experiment introduces competition for the initial inspection into a market with endogenous pricing and consumer search. Sellers learn their auction-assigned positions before pricing, recall remains available, and the treatment varies the monetary cost of the second inspection. Observing the auction and product market together permits a welfare comparison that includes the payment for prominence, alongside the prices consumers pay and the search costs they incur.

The remainder of the paper is organized as follows. Section 2 presents the model, the buyer's search cutoffs, and the pricing and bidding equilibria. Section 3 describes the experimental design and implementation. Section 4 presents the welfare findings and then examines bidding, pricing, and search behavior. Section 5 concludes. Appendix A contains the proofs and conditional equilibrium benchmarks, and Appendix B contains the experimental instructions.

2 Model

2.1 Environment and timing

There are two ex ante identical sellers and one buyer with unit demand. Each seller offers one product at zero marginal cost. The buyer's match values are independent across sellers, with

$$\Pr(v_k = V) = q, \quad \Pr(v_k = 0) = 1 - q, \quad k = 1, 2, \quad (1)$$

where $V > 0$ and $0 < q < 1$. All players know the distribution, but sellers do not observe the buyer's match values when bidding or pricing. The buyer can exit for a payoff of zero. The cost of inspecting the second product satisfies $0 < s < qV$.

The game has three stages.

1. *Auction for prominence.* Sellers simultaneously submit nonnegative bids in a second-price auction. The higher bidder receives the top position and pays the losing bid, $b^{(2)}$. The other seller receives the second position and pays nothing. Equal bids are resolved by an equal-probability draw.
2. *Pricing.* After learning their positions and the position payment, sellers simultaneously post prices $p_1, p_2 \geq 0$. Subscripts henceforth identify positions: seller 1 is the top seller

and seller 2 is the second seller.

3. *Search and purchase.* The buyer observes both prices and inspects the top product for free, learning v_1 . She may pay s to inspect the second product and learn v_2 . At either decision stage, she may purchase an inspected product or exit. Recall of the top product is free, but purchase from seller 2 requires inspection.

Let $x \in \{0, 1\}$ denote inspection of the second product and $a \in \{0, 1, 2\}$ the purchase decision, with $a = 0$ denoting exit. The buyer's payoff is

$$u(a, x) = \sum_{k=1}^2 (v_k - p_k) \cdot \mathbf{1}_{\{a=k\}} - s \cdot x. \quad (2)$$

Product-market revenues, net seller profits, and platform revenue are

$$\Pi_k = p_k \cdot \mathbf{1}_{\{a=k\}}, \quad \pi_1 = \Pi_1 - b^{(2)}, \quad \pi_2 = \Pi_2, \quad \text{PR} = b^{(2)}. \quad (3)$$

The position payment is sunk when sellers choose prices. I solve backward from the buyer's search and purchase decisions.

2.2 Buyer search and seller demand

After observing (v_1, p_1, p_2) , the buyer compares the expected gain from inspecting the second product with its cost. With a bad top match, search offers a chance of finding a product worth buying. With a good top match, it offers a chance of purchasing a suitable product at a lower price. The following rules specify optimal behavior at indifference.

- A1.** With $v_1 = V$, the buyer stops when indifferent between searching and stopping.
- A2.** With $v_1 = 0$, the buyer searches when indifferent between searching and stopping.
- A3.** Among available terminal actions yielding the same highest surplus, the buyer prefers purchase from seller 1 to purchase from seller 2, and purchase from seller 2 to exit.

Lemma 1 (Optimal search cutoffs). *For $0 \leq p_1, p_2 \leq V$, the buyer's optimal search decision satisfies*

$$x^*(0, p_1, p_2) = 1 \iff p_2 \leq V - \frac{s}{q}, \quad (4)$$

$$x^*(V, p_1, p_2) = 1 \iff p_1 - p_2 > \frac{s}{q}. \quad (5)$$

After any inspection, the buyer selects the available product with the largest nonnegative purchase surplus, following A3 at ties.

Proof. See Appendix A.1, which also specifies optimal behavior after prices outside $[0, V]$. $\square$

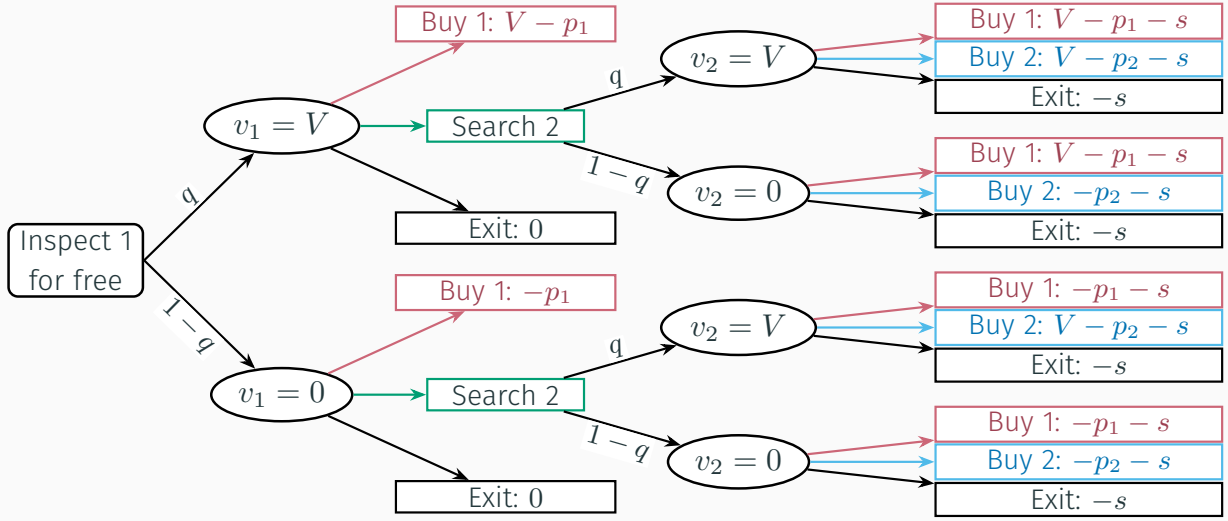


Figure 1: The buyer's search and purchase decisions

The buyer observes both prices and inspects seller 1 for free. Inspection of seller 2 costs s . The tree displays the available actions and their payoffs.

For $0 < p_1 \leq V$ and $0 < p_2 \leq V - \frac{s}{q}$, let $D_k(p_1, p_2)$ be seller k 's sale probability. The search cutoffs imply

	$D_1(p_1, p_2)$	$D_2(p_1, p_2)$	
$p_1 - p_2 > \frac{s}{q}$	$q(1 - q)$	q	
$p_1 - p_2 \leq \frac{s}{q}$	q	$q(1 - q)$	(6)

When a good-top-match buyer stops, seller 1 serves all such buyers, while seller 2 sells only when the first match is bad and the second is good. When the price gap induces search, seller 2 also attracts buyers with two good matches. Seller 1 then sells only when the first match is good and the second is bad. Figures 2 and 3 show the realized paths in the two cases. At a zero price, the purchase tie rule can change unit demand, but the zero-price seller's revenue is always zero.

If $p_1 - p_2 > \frac{s}{q}$, a buyer with $v_1 = V$ **searches** (assuming $0 < p_1 \leq V$, $0 < p_2 \leq V - \frac{s}{q}$)

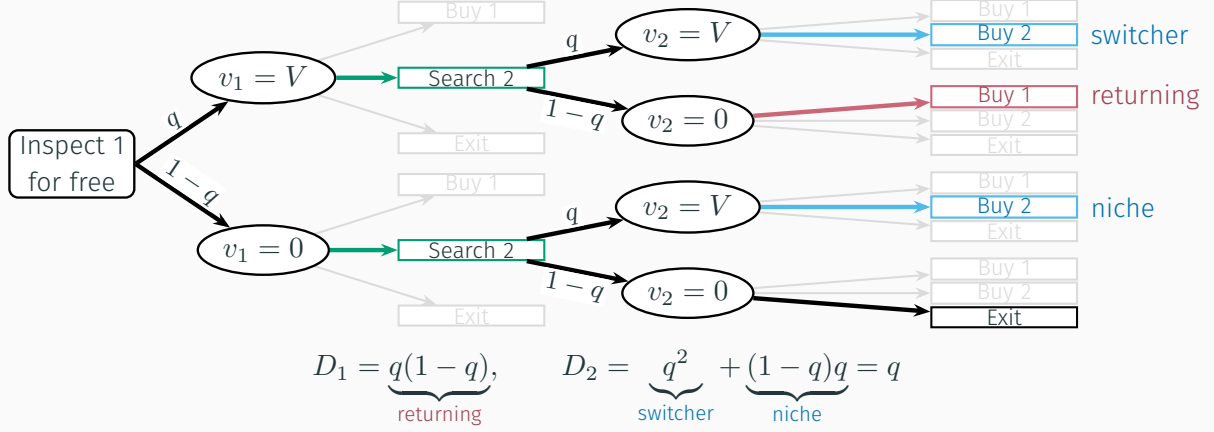


Figure 2: Search activated after a good top match

The price gap exceeds $\frac{s}{q}$. Active paths determine seller demand, while unrealized paths are shown in gray.

If $p_1 - p_2 \leq \frac{s}{q}$, a buyer with $v_1 = V$ does **not search** (assuming $0 < p_1 \leq V$, $0 < p_2 \leq V - \frac{s}{q}$)

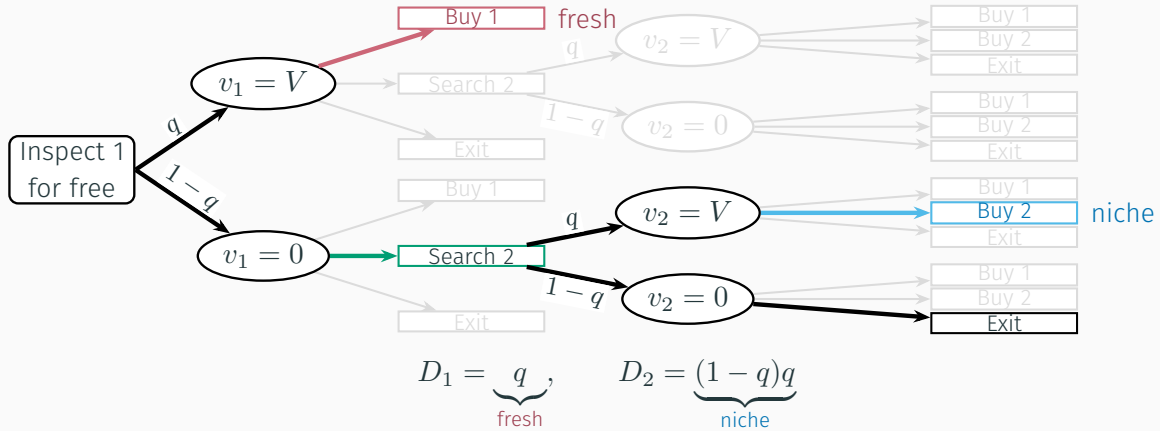


Figure 3: Search deterred after a good top match

The price gap is at most $\frac{s}{q}$. A good-top-match buyer stops, while a bad-top-match buyer inspects seller 2 within the displayed price range.

2.3 Pricing equilibrium

Proposition 1 (No pure-strategy Nash equilibrium). *Under A1–A3 and $0 < s < qV$, the pricing subgame has no pure-strategy Nash equilibrium.*

Proof. See Appendix A.2. □

On a constant-demand segment, each seller benefits from raising its price. At the cutoff, seller 2 can instead gain demand by slightly undercutting. These opposing incentives motivate mixed pricing strategies.

At deviation prices outside the domain in (6), demand follows the buyer's optimal strategy.

Definition 1 (Mixed-strategy pricing equilibrium). We seek distribution functions $F_1(p_1)$ and $F_2(p_2)$ with connected supports $[\underline{p}_1, \bar{p}_1]$ and $[\underline{p}_2, \bar{p}_2]$ such that

$$p_1^* \in \arg \max_{p_1 \in [0, \infty)} p_1 \mathbb{E}_{F_2}[D_1(p_1, p_2)], \quad \forall p_1^* \in [\underline{p}_1, \bar{p}_1], \quad (7)$$

$$p_2^* \in \arg \max_{p_2 \in [0, \infty)} p_2 \mathbb{E}_{F_1}[D_2(p_1, p_2)], \quad \forall p_2^* \in [\underline{p}_2, \bar{p}_2]. \quad (8)$$

Theorem 1 (A mixed-strategy pricing equilibrium). *Under A1–A3 and $0 < s < qV$, the following independent distributions constitute an equilibrium of the pricing subgame:*

$$F_1(p_1) = \frac{1}{q} \left[1 - \frac{(1-q) \left(V - \frac{s}{q} \right)}{p_1 - \frac{s}{q}} \right], \quad \underline{p}_1 \leq p_1 \leq \bar{p}_1, \quad (9)$$

$$F_2(p_2) = \frac{1}{q} \left[1 - \frac{(1-q)V + s}{p_2 + \frac{s}{q}} \right], \quad \underline{p}_2 \leq p_2 < \bar{p}_2, \quad (10)$$

where

$$\begin{aligned} \underline{p}_1 &= (1-q)V + s, & \bar{p}_1 &= V, \\ \underline{p}_2 &= (1-q) \left(V - \frac{s}{q} \right), & \bar{p}_2 &= V - \frac{s}{q}. \end{aligned} \quad (11)$$

Both distributions are zero below their lower bounds and one at or above their upper bounds. Seller 1's distribution is atomless. Seller 2 places mass $\frac{s}{qV}$ at $\bar{p}_2$.

Proof. Appendix A.3 verifies both optimality conditions against all nonnegative price deviations. □

The support bounds satisfy

$$\underline{p}_1 - \underline{p}_2 = \bar{p}_1 - \bar{p}_2 = \frac{s}{q}. \quad (12)$$

At the pair of upper bounds, a good-top-match buyer stops and a bad-top-match buyer searches. The tie rules thus preserve each seller's demand at its upper price. With the

experimental parameters $V = 100$ and $q = \frac{1}{2}$, the supports are $[70, 100]$ and $[30, 60]$ at $s = 20$, and $[90, 100]$ and $[10, 20]$ at $s = 40$. Top prices therefore first-order stochastically dominate second prices in both treatments.

2.4 Bidding for prominence

Theorem 2 (Symmetric bidding equilibrium). *Suppose the sellers use the pricing equilibrium in Theorem 1 after every auction outcome. Their expected product-market revenues, which are the continuation values before subtracting any position payment, are*

$$\mathbb{E}[\Pi_1] = q[(1 - q)V + s], \quad \mathbb{E}[\Pi_2] = q(1 - q) \left(V - \frac{s}{q} \right). \quad (13)$$

The value of prominence is their difference,

$$\mathbb{E}[\Pi_1] - \mathbb{E}[\Pi_2] = s. \quad (14)$$

The symmetric bids $b_1 = b_2 = b^ = s$, together with these pricing strategies and optimal buyer behavior, constitute a subgame-perfect equilibrium. Platform revenue and expected net seller profits are*

$$\text{PR} = s, \quad \mathbb{E}[\pi_1] = \mathbb{E}[\pi_2] = (1 - q)(qV - s). \quad (15)$$

Proof. See Appendix A.4. □

The auction payment absorbs the top seller's expected revenue advantage. At $(V, q) = (100, \frac{1}{2})$, expected revenues are (35, 15) when $s = 20$ and (45, 5) when $s = 40$. Expected net profits are therefore 15 per seller in the first treatment and 5 in the second.

2.5 Welfare and empirical benchmarks

Consumer surplus is the buyer's payoff. Total surplus includes both sellers and the platform:

$$\text{CS} = \sum_{k=1}^2 (v_k - p_k) \cdot \mathbf{1}_{\{a=k\}} - s \cdot x, \quad (16)$$

$$\text{TS} = \text{CS} + \pi_1 + \pi_2 + \text{PR} = \sum_{k=1}^2 v_k \cdot \mathbf{1}_{\{a=k\}} - s \cdot x. \quad (17)$$

Prices and position payments are transfers. Search uses real resources, so total surplus depends on the allocation and the search decision. For each observed market, I calculate the equilibrium benchmark conditional on its realized match values, averaging over the equilibrium price distributions and the implied buyer choices. Appendix A.6 gives these benchmarks. This comparison holds the realized matches fixed when evaluating departures from equilibrium behavior.

3 Experimental Design

The experiment implements the market described above. Each market consists of two sellers and one buyer. Each product independently has a good match value of $V = 100$ points with probability $q = 0.5$, and a bad match value of zero otherwise. I vary the cost of inspecting the second seller between sessions: $s = 20$ in the low-cost treatment and $s = 40$ in the high-cost treatment. All other market rules are the same. Subjects participate in only one treatment.

The three-stage sequence follows Section 2.1: sellers bid for positions, learn their positions and set prices, and the buyer chooses whether to inspect and where to purchase. Bids and prices are restricted to whole points from 0 to 100. The continuous-price equilibrium serves as the theoretical benchmark for this discrete implementation. Both prices and the top match are visible before inspection; the second product can be purchased only after inspection, and recall of the top product is free.

The treatment changes the value of prominence through the buyer’s search decision. With a bad top match, the cutoff for the second seller’s price falls from 60 to 20 points as the search cost rises. With a good top match, the price reduction needed to justify inspection rises from 40 to 80 points. The equilibrium bid consequently rises from 20 to 40 points. This comparison tests whether a greater search disadvantage for the second seller translates into higher auction payments and how the resulting bidding, pricing, and search decisions affect the distribution of surplus.

I conducted six computerized sessions at the University of Virginia’s Veconlab, three per treatment. There were 42 subjects in the low-cost treatment and 45 in the high-cost treatment. Five sessions had 15 subjects; one low-cost session had 12. Roles were assigned at the beginning and remained fixed. In each round, the computer randomly and anonymously rematched subjects into groups of two sellers and one buyer, and drew new match values independently. Each subject completed 30 incentivized rounds. Thus, the sample contains 420 low-cost and 450 high-cost market-round observations, generated by 14 and 15 buyers, respectively. Table 1 summarizes the design.

Subjects received written instructions and answered three comprehension questions before completing two unpaid practice rounds. At the end of each incentivized round, they received feedback on the realized history of their market and their own payoff. Feedback concerned information revealed during play; the second match value was not disclosed when the buyer had not inspected that product.

Every round began with a 200-point balance, reset independently of earlier earnings. This balance covered possible losses from inspection and position payments. One of the 30 incentivized rounds was randomly selected for payment. The selected round’s balance plus payoff was converted at 10 points per dollar and added to a \$10 show-up fee. The welfare measures below exclude the initial balance and show-up payment: they measure surplus generated by trading and search within the market. Appendix B reproduces the retained

Table 1: Experimental design and sample

	$s = 20$	$s = 40$
Sessions	3	3
Subjects	42	45
Sellers	28	30
Buyers	14	15
Incentivized rounds per subject	30	30
Markets per round, across sessions	14	15
Market-round observations	420	450

Roles remain fixed; markets are randomly rematched within sessions each round. Each market contains two sellers and one buyer. Match values are independently redrawn each round.

instructions, documenting the decision sequence, information, and payment rules.

Random rematching limits repeated interactions with an identified opponent but does not make the 870 markets independent observations. Subjects take part repeatedly, sellers and buyers share histories through rematching, and treatment varies across six sessions. The results therefore distinguish pooled market comparisons from the corresponding participant-history and session-level checks. These checks describe the sensitivity of inference to the unit of analysis; the session sample remains small.

4 Experimental Results

I first compare welfare outcomes with equilibrium, then examine the bidding, pricing, and search decisions behind these outcomes. The analysis uses all 30 incentivized rounds. Unless stated otherwise, reported means pool the 420 low-cost and 450 high-cost markets, giving each market equal weight.

4.1 Welfare

For each market, consumer surplus is the value of the purchased product minus its price and any inspection cost. Platform revenue is the losing bid paid by the top seller. Combined seller profits equal the two sellers' sales revenues less that payment. Equation (17) gives the accounting identity: prices and position payments redistribute surplus, while allocation and inspection determine its total. An inspection cost is counted even when the buyer exits or returns to the top seller.

The equilibrium comparison uses the same match draws as the experiment. For each market, I calculate the expected equilibrium outcome conditional on its realized pair (v_1, v_2) , integrating over the equilibrium price distributions, and then average these conditional predictions. Buyers in this calculation retain the model's information structure: they do not

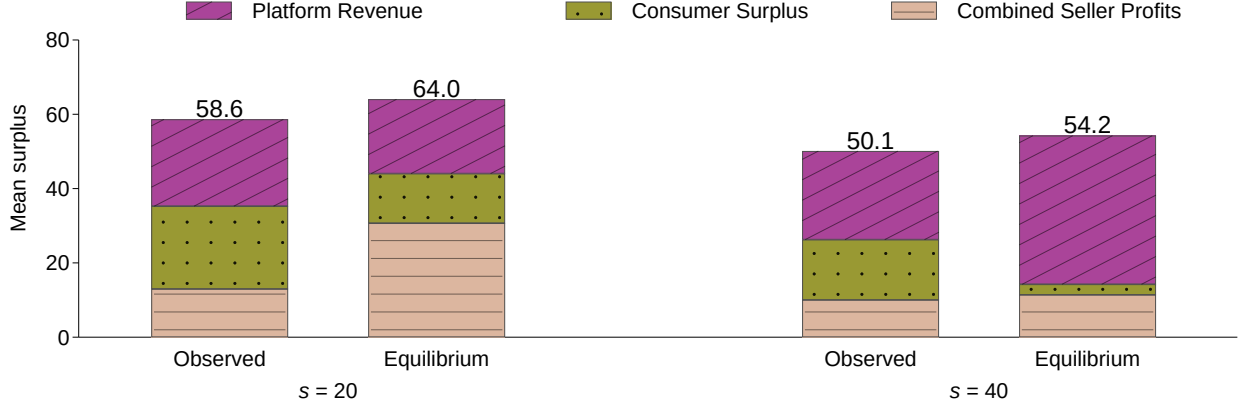


Figure 4: The level and distribution of surplus

Means in experimental points. Each bar is total surplus, divided into platform revenue, consumer surplus, and combined seller profits. Equilibrium outcomes condition on the observed match draws and integrate over equilibrium prices. Values above the bars report total surplus; Table 2 reports each component and comparison. The figure displays means without confidence intervals.

Table 2: Welfare outcomes and equilibrium benchmarks

Outcome	$s = 20$			$s = 40$			Across costs
	Obs.	Eq.	p	Obs.	Eq.	p	p
Total surplus	58.62	64.03	< 0.001	50.09	54.23	0.004	0.013
Platform revenue	23.35	20.00	< 0.001	23.83	40.00	< 0.001	0.718
Consumer surplus	22.29	13.34	< 0.001	16.23	2.84	< 0.001	0.002
Combined seller profits	12.98	30.69	< 0.001	10.03	11.39	0.358	0.233

Means are points per market over all 30 rounds ($N = 420$ at $s = 20$ and $N = 450$ at $s = 40$). Equilibrium values condition on the realized match draws and integrate equilibrium price mixing. Within-cost p -values are paired observed-versus-equilibrium t tests; across-cost values are Welch tests of observed means. All tests are two-sided and unadjusted. These market-level tests treat repeated markets as independent. Seller profits subtract the position payment.

learn v_2 before deciding whether to inspect. Conditioning adjusts the comparison for the finite sample of match draws; it does not give the benchmark buyer additional information. In particular, the equilibrium platform payment remains s in every market. The observed-draw adjustment can make the two positions' sample-average equilibrium profits differ even though their ex ante equilibrium profits are equal.

Figure 4 and Table 2 report the four welfare components. Within each cost, observed and equilibrium outcomes are paired market by market. Across costs, the reported comparisons use two-sided Welch tests. These pooled tests are descriptive references: their standard errors treat market observations as independent despite repeated subjects and rematching. The pooled sample size is therefore not the number of independent treatment assignments. All reported p -values are unadjusted for multiple comparisons.

Total surplus. Observed total surplus falls from 58.62 at $s = 20$ to 50.09 at $s = 40$, a decline of 8.53 points ($p = 0.013$). The draw-conditional equilibrium benchmarks are 64.03 and 54.23, a decline of 9.80 points. Thus, the observed change has the predicted direction, while total surplus is below equilibrium in both treatments. The observed-minus-equilibrium gaps are -5.41 at the low cost and -4.15 at the high cost (paired $p < 0.001$ and $p = 0.004$).

The accounting in (17) clarifies these gaps. The mean allocated match value is 66.67 at the low cost and 60.22 at the high cost, compared with benchmarks of 76.90 and 75.11. Observed search expenditure is also lower: 8.05 versus 12.87, and 10.13 versus 20.88. Fewer resources are spent on inspection, but the reduction in realized allocation value is larger. These are accounting differences between observed and equilibrium outcomes; they do not identify separate causal effects of pricing and search behavior.

Platform revenue. The predicted 20-point increase in auction revenue does not appear in the pooled means. Platform revenue is 23.35 at $s = 20$ and 23.83 at $s = 40$, an increase of 0.48 points ($p = 0.718$). The 95% Welch interval for this increase is $[-2.11, 3.07]$. This estimate is small relative to the predicted increase of 20 points, although nonrejection of equal observed means does not establish equality. Revenue exceeds the low-cost equilibrium benchmark by 3.35 points and falls below the high-cost benchmark by 16.17 points (both market-level $p < 0.001$). The relevant auction statistic is the losing bid, which determines the payment, rather than the average submitted bid.

Consumer surplus. Consumers receive more surplus than predicted in both treatments. Observed consumer surplus is 22.29 at $s = 20$, against a benchmark of 13.34, and 16.23 at $s = 40$, against 2.84 (both paired $p < 0.001$). It falls by 6.06 points across treatments ($p = 0.002$), compared with a predicted decline of 10.50 points.

The proportional comparison is sharper: observed consumer surplus falls by 27.2%, whereas its draw-conditional equilibrium benchmark falls by 78.7%. These percentages are changes in treatment means, not averages of individual percentage changes. A delta-method comparison, retaining the covariance between observed and equilibrium surplus within each treatment, gives $p < 0.001$ for the 51.5-percentage-point difference. This is an asymptotic market-level reference and inherits the dependence qualification above. The higher level of consumer surplus is compatible with lower total surplus: consumers make lower payments and incur less search expenditure, while some valuable purchases fail to occur.

Combined seller profits. The two sellers together earn 12.98 after the position payment at $s = 20$, well below the equilibrium benchmark of 30.69 (paired $p < 0.001$). At $s = 40$, their mean profit is 10.03 against a benchmark of 11.39; the 1.36-point shortfall is comparatively small and is not statistically detected ($p = 0.358$). The across-treatment difference is also imprecisely estimated ($p = 0.233$). Sellers retain 22.1% and 20.0% of observed total surplus,

Table 3: Bids, posted prices, and seller revenues

Outcome	$s = 20$			$s = 40$		
	Obs.	Eq.	p	Obs.	Eq.	p
Top price	61.83	81.59	< 0.001	66.07	93.86	< 0.001
Second price	48.90	49.93	0.196	39.11	18.96	< 0.001
Pooled bid	34.03	20.00	< 0.001	38.79	40.00	0.221
Winning bid	44.71	20.00	< 0.001	53.76	40.00	< 0.001
Losing bid / payment	23.35	20.00	< 0.001	23.83	40.00	< 0.001
Top gross revenue	27.67	34.17	< 0.001	30.89	46.73	< 0.001
Second gross revenue	8.66	16.52	< 0.001	2.97	4.66	< 0.001
Combined gross revenue	36.33	50.69	< 0.001	33.86	51.39	< 0.001

All-round means in points. Pooled bids average the two submitted bids in each market. Price predictions are the analytical means of the equilibrium price distributions; revenue predictions additionally condition on the observed match draws. A seller’s gross revenue is the posted price if it makes a sale and zero otherwise. The losing bid is exactly the top seller’s payment to the platform. Two-sided p -values compare observed and equilibrium outcomes: one-sample tests against analytical price/bid means and paired tests for revenues. Market observations are treated as independent, and p -values are unadjusted.

respectively. These shares are ratios of treatment means, not averages of market-specific ratios.

The proximity to equilibrium profit at the high search cost does not imply that seller behavior or gross revenues are close to equilibrium. As Table 3 shows, combined sales revenue is 36.33 rather than 50.69 at the low cost and 33.86 rather than 51.39 at the high cost. Revenue at each position is below its draw-conditional benchmark at both costs (all paired market-level $p < 0.001$). The profit identity explains the different net gaps. At $s = 20$, a revenue shortfall of 14.36 and an auction payment 3.35 above equilibrium together produce the 17.71-point profit shortfall. At $s = 40$, the revenue shortfall of 17.53 is largely offset by an auction payment 16.17 below equilibrium, leaving a net shortfall of 1.36. This is an exact accounting relation, not a decomposition into independently identified behavioral effects.

4.2 Bidding, Pricing, and Search

Bidding and the platform payment. Table 3 separates pooled, winning, and losing bids. The mean submitted bid rises from 34.03 to 38.79, an increase of 4.76 points. At the market level, this increase differs from zero and is smaller than the predicted increase of 20 (both $p < 0.001$). For this test, the pooled-bid observation is the average of the two bids within a market, so the two bids are not treated as separate observations. The pooled mean is above the equilibrium bid of 20 in the low-cost treatment ($p < 0.001$); its difference from 40 in the high-cost treatment is not detected ($p = 0.221$).

Winning bids rise from 44.71 to 53.76, while losing bids remain near 24. Since the top seller pays the losing bid, the increase in winning bids does not itself increase platform revenue.

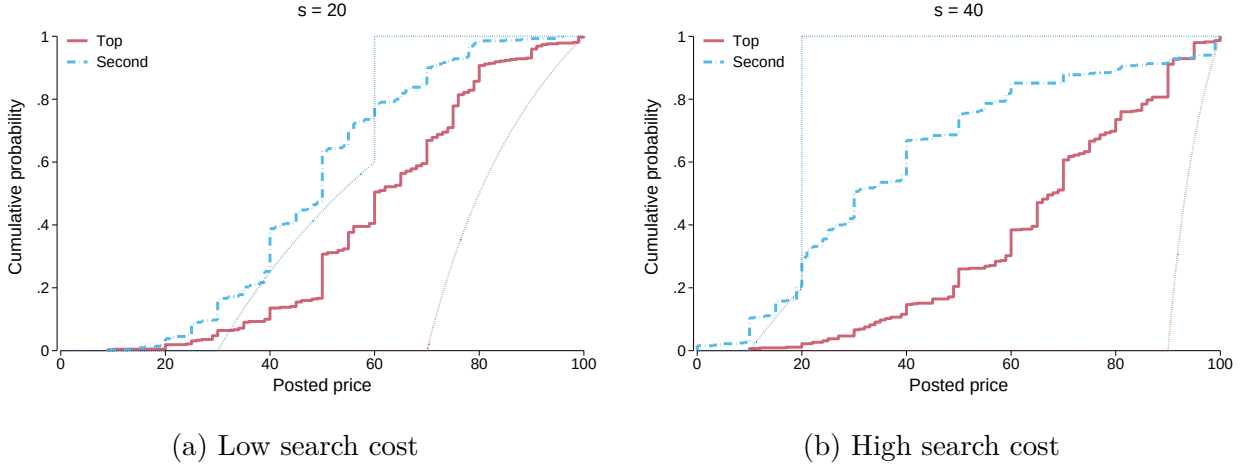


Figure 5: Observed and equilibrium price distributions

All 30 rounds. Red and blue identify the top and second sellers, respectively. Dotted curves show the equilibrium CDFs; solid and dash-dot curves show the corresponding empirical CDFs. These distributions concern posted prices, whether or not a sale occurs.

The high-cost pooled mean also conceals substantial differences between winning and losing bids: a mean near 40 need not imply that individual bids concentrate near the equilibrium bid. The across-cost pooled-bid comparison is less precise when bids are first averaged by seller or session ($p = 0.383$ and $p = 0.663$, respectively). The stronger pooled-market rejection therefore should not be read as evidence from hundreds of independent sellers.

Posted prices and the opportunity to trade. Figure 5 compares observed and equilibrium price distributions. Top sellers post substantially lower prices than predicted: their means are 61.83 and 66.07, compared with equilibrium means of 81.59 and 93.86 (one-sample market-level $p < 0.001$ at both costs). The second seller’s mean is 48.90 against 49.93 at the low cost ($p = 0.196$), but 39.11 against 18.96 at the high cost ($p < 0.001$). These tests compare observed prices with the fixed analytical equilibrium means. The posted-price distributions also extend well beyond the equilibrium supports. Top-price standard deviations are 17.48 and 20.27, against predicted values of 8.39 and 2.80. Second-price standard deviations are 16.30 and 25.36, against 10.57 and 2.44. These dispersion comparisons are descriptive.

Top prices exceed second prices in 66.7% of low-cost markets and 81.6% of high-cost markets. The corresponding mean gaps are 12.93 and 26.96, below the predicted 31.65 and 74.90. These paired comparisons establish neither a uniform price ordering nor first-order stochastic dominance: the empirical cumulative distributions cross. More importantly, an average price gap does not determine inspection after a bad top match. In that case, the relevant comparison is the second price with $100 - 2s$, independently of the top price.

The second seller exceeds that bad-match inspection cutoff in 21.4% of all low-cost markets and 70.2% of all high-cost markets. Among buyers who actually receive a bad top match,

Table 4: Search decisions relative to the optimal cutoff

Top match	s	Strict cases	Compliant	Under-search	Over-search	Ties
Good	20	199	90.5%	9	10	5
Good	40	233	89.7%	1	23	1
Bad	20	210	80.5%	32	9	6
Bad	40	195	64.6%	20	49	21

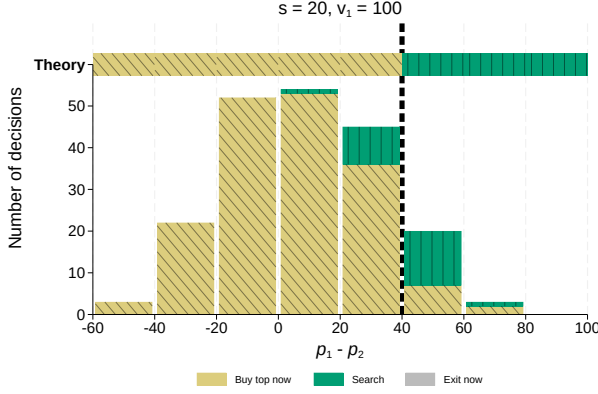
Compliance classifies the inspection decision at observed prices. Exact inspection ties are excluded from the strict cases and reported separately. Under-search means stopping when inspection has strictly positive net expected value; over-search means inspecting when stopping is strictly preferable. Counts of errors are not conditional error rates. Purchase optimality is a separate decision.

the second price lies strictly above the cutoff in 46 of 216 low-cost observations and 143 of 216 high-cost observations. At those prices, stopping is optimal, so a buyer who follows the cutoff does not discover or buy the second product even when it would be a good match. This limits second-seller sales. Lower top prices, meanwhile, make top purchases cheaper for consumers and reduce revenue per top sale. These price patterns and the search choices below are consistent with the observed consumer-surplus and revenue gaps; their separate causal contributions are not identified by the treatment comparison.

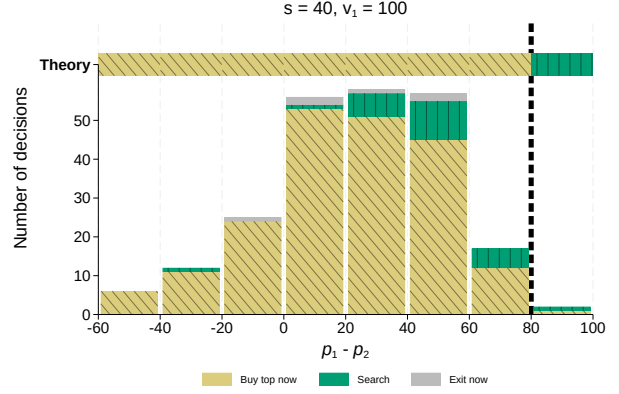
Search decisions. I evaluate inspection choices at the prices subjects actually observe. With a good top match, inspection is strictly optimal when $p_1 - p_2 > 2s$; with a bad top match, it is strictly optimal when $p_2 < 100 - 2s$. Figure 6 displays decisions around these cutoffs. Table 4 excludes exact inspection ties when calculating compliance and reports their counts separately. This avoids classifying an indifferent buyer’s inspection choice as a strict error. The equilibrium tie convention remains relevant to the theoretical benchmark, but the empirical accuracy measure does not require subjects to follow that convention.

After a good top match, 90.5% of low-cost and 89.7% of high-cost decisions satisfy the strict inspection rule. These high rates partly reflect the prices buyers face. In the high-cost treatment, 231 of the 233 strict cases call for stopping, leaving only two cases in which inspection is strictly preferable. Overall compliance therefore provides limited evidence about behavior on both sides of that cutoff. After a bad top match, compliance is lower: 80.5% at $s = 20$ and 64.6% at $s = 40$.

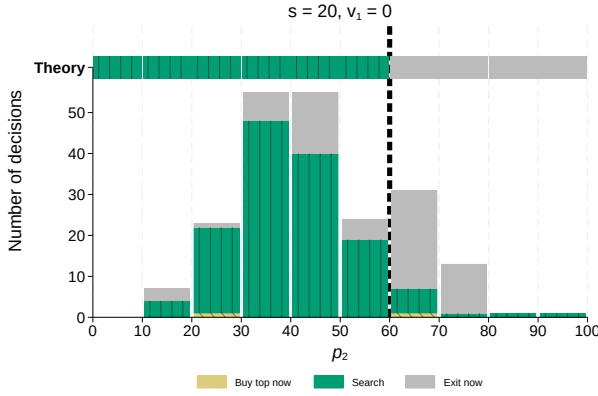
Errors occur in both directions. Low-cost buyers with a bad top match fail to inspect in 32 of the 164 cases in which inspection is strictly optimal, and inspect in 9 of the 46 cases in which stopping is optimal. At the high cost, the corresponding counts are 20 of 52 and 49 of 143. The conditional under-search rates are therefore 19.5% and 38.5%, and the over-search rates are 19.6% and 34.3%. Under-search can prevent a valuable second purchase; over-search incurs an inspection cost when it is not justified by the buyer’s expected private gain. Search accuracy is distinct from purchase accuracy after the available products have been revealed.



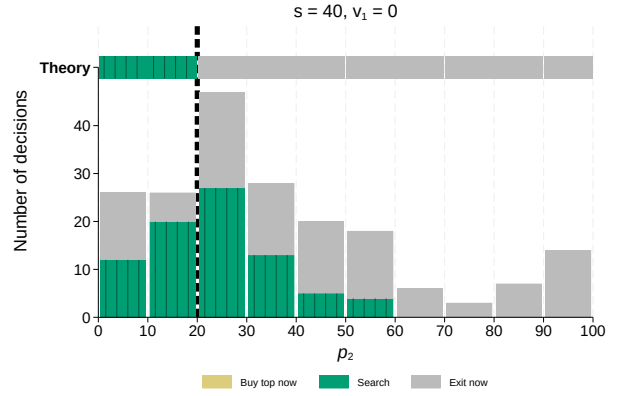
(a) Good top match, low cost



(b) Good top match, high cost



(c) Bad top match, low cost



(d) Bad top match, high cost

Figure 6: Search decisions at observed prices

All 30 rounds. Bar heights show the frequency of observed prices or price differences; segments distinguish buying from the top seller, inspecting the second seller, and exit at the first decision. The price difference governs inspection after a good top match; the second price governs it after a bad top match. The strip marked Theory indicates the optimal action in each price region. Cutoff ties are shown in the figures but excluded from the strict-compliance rates in Table 4.

The decline in bad-match compliance is descriptive evidence of a different response to the high-cost environment. The session-level exact Wilcoxon rank-sum comparison gives $p = 0.100$ for bad top matches and $p = 1.000$ for good top matches. With three sessions per treatment, these comparisons are coarse and have limited power. They do not establish that the two treatments generate equal search behavior.

Purchase choices are much closer to the payoff-maximizing choice within the buyer’s chosen information set: there are four purchase errors among 420 low-cost markets and six among 450 high-cost markets. The principal departures therefore concern how sellers bid and price and whether buyers inspect the second product, rather than which inspected product buyers choose.

Sensitivity to the unit of analysis. The six-session design limits inference about treatment effects. At the session level, the declines in total surplus and consumer surplus give $p = 0.063$ and $p = 0.066$, compared with $p = 0.013$ and $p = 0.002$ in the pooled-market comparisons. Several observed-versus-equilibrium conclusions also become less precise. The results establish the magnitudes and accounting relationships observed in these experimental markets; their statistical precision depends on the treatment of repeated play and common session conditions.

5 Conclusion

This paper studies a market in which sellers compete for prominence before setting prices and consumers decide how far to search. The model links the value of the prominent position to the pricing and search decisions that follow its allocation. A laboratory experiment varies the cost of inspecting the non-prominent product and observes how surplus is divided among consumers, sellers, and the platform.

The welfare results have three main features. Platform auction revenue changes little across search costs, reflecting the weak response of losing bids. Consumer surplus exceeds its equilibrium benchmark at both costs and falls proportionally less than predicted as the cost increases. Sellers jointly retain only about one fifth of total surplus after paying for prominence. Lower top prices benefit buyers who purchase from the prominent seller but reduce seller revenue per sale. Second prices above the search cutoff can discourage further inspection after a bad initial match and lead to lost sales. Combined sales revenue is below equilibrium in both treatments, and position payments take a substantial share of that revenue. At the high search cost, payments below equilibrium offset much of the revenue shortfall, explaining why net profits are closer to the benchmark there.

The experiment separates the search-cost advantage of prominence from information that rankings may convey in the field. It also shows why bids and product-market outcomes

should be studied together. Auction payments depend on sellers' valuations of prominence, while those valuations depend on anticipated pricing and consumer search. An auction model that takes position values as given can describe payment incentives, but cannot alone explain how a change in consumer search friction changes those values. Conversely, a model that assigns prominence without payment misses the transfer of seller revenue to the platform.

These findings suggest several directions for further work. One concerns platforms' incentives to change search costs. In the experiment, a lower cost improves consumer and total surplus without a statistically detectable decline in auction revenue. This comparison does not establish an optimal search cost for a platform: investment in search technology may be costly, and transaction commissions, participation, and other revenue sources may also respond. An endogenous platform decision would need to incorporate these margins rather than equate platform profit with auction revenue alone.

A second question is how prominence should be allocated. Comparing the present auction with random allocation or a first-price auction would show how outcomes depend on the allocation and payment mechanisms. With more sellers and several degrees of prominence, the allocation could also affect which products enter consumers' consideration sets. Seller differences in quality, costs, or match distributions would then matter for whether willingness to pay selects the socially preferred search order. The two identical sellers in this paper allow the value of an inspection-cost advantage to be isolated; they do not settle that broader allocation question.

A third question concerns the product information that sellers can display to consumers without further inspection. Here, prices are visible but match values must be learned through search. Other information structures may change how sellers direct attention and how buyers choose which products to inspect. Studying price visibility alongside product characteristics, reviews, and platform disclosure rules would help connect competition for prominence to the design of the search environment itself.

References

- ABRAMS, E., M. SEFTON, AND A. YAVAS (2000): “An Experimental Comparison of Two Search Models,” *Economic Theory*, 16, 735–749.
- ANDERSON, S. P. AND R. RENAULT (1999): “Pricing, Product Diversity, and Search Costs: A Bertrand–Chamberlin–Diamond Model,” *The RAND Journal of Economics*, 30, 719–735.
- (2026): “Search Direction: Position Externalities and Position Auction Bias,” *The Review of Economic Studies*, 93, 763–797.
- ARMSTRONG, M. (2017): “Ordered Consumer Search,” *Journal of the European Economic Association*, 15, 989–1024.
- ARMSTRONG, M., J. VICKERS, AND J. ZHOU (2009): “Prominence and Consumer Search,” *The RAND Journal of Economics*, 40, 209–233.
- ARMSTRONG, M. AND J. ZHOU (2011): “Paying for Prominence,” *The Economic Journal*, 121, F368–F395.
- ATHEY, S. AND G. ELLISON (2011): “Position Auctions with Consumer Search,” *The Quarterly Journal of Economics*, 126, 1213–1270.
- BROWN, A. L., A. VIRIYAVIPART, AND X. WANG (2018): “Search Deterrence in Experimental Consumer Goods Markets,” *European Economic Review*, 104, 167–184.
- CASON, T. N. AND D. FRIEDMAN (2003): “Buyer Search and Price Dispersion: A Laboratory Study,” *Journal of Economic Theory*, 112, 232–260.
- CHEN, Y. AND C. HE (2011): “Paid Placement: Advertising and Search on the Internet,” *The Economic Journal*, 121, F309–F328.
- CHOI, M., A. Y. DAI, AND K. KIM (2018): “Consumer Search and Price Competition,” *Econometrica*, 86, 1257–1281.
- DAVIS, D. D. AND C. A. HOLT (1996): “Consumer Search Costs and Market Performance,” *Economic Inquiry*, 34, 133–151.
- DIAMOND, P. A. (1971): “A Model of Price Adjustment,” *Journal of Economic Theory*, 3, 156–168.
- DING, Y. AND T. ZHANG (2018): “Price-Directed Consumer Search,” *International Journal of Industrial Organization*, 58, 106–135.
- EDELMAN, B., M. OSTROVSKY, AND M. SCHWARZ (2007): “Internet Advertising and the Generalized Second-Price Auction: Selling Billions of Dollars Worth of Keywords,” *American Economic Review*, 97, 242–259.
- OBRADOVITS, M. AND P. PLAICKNER (2023): “Price-Directed Search, Product Differentiation and Competition,” *Review of Industrial Organization*, 63, 317–348.
- URSU, R. M. (2018): “The Power of Rankings: Quantifying the Effect of Rankings on Online Consumer Search and Purchase Decisions,” *Marketing Science*, 37, 530–552.
- VARIAN, H. R. (2007): “Position Auctions,” *International Journal of Industrial Organization*, 25, 1163–1178.
- WOLINSKY, A. (1986): “True Monopolistic Competition as a Result of Imperfect Information,” *The Quarterly Journal of Economics*, 101, 493–511.
- XU, L., J. CHEN, AND A. B. WHINSTON (2011): “Price Competition and Endogenous Valuation in Search Advertising,” *Journal of Marketing Research*, 48, 566–586.

ZHOU, J. (2011): “Ordered Search in Differentiated Markets,” *International Journal of Industrial Organization*, 29, 253–262.

A Proofs

A.1 Proof of Lemma 1

At any $p_1, p_2 \geq 0$, stopping gives $\max\{0, v_1 - p_1\}$. The expected gain from inspection before paying its cost is

$$q \max\{0, v_1 - p_1, V - p_2\} + (1 - q) \max\{0, v_1 - p_1, -p_2\} - \max\{0, v_1 - p_1\}. \quad (18)$$

The buyer searches when this gain exceeds s and stops when it is smaller. A1 and A2 resolve equality. Once inspection is complete, the search cost is sunk, and A3 selects a surplus-maximizing terminal action.

For $0 \leq p_1, p_2 \leq V$ and $v_1 = 0$, the gain is $q(V - p_2)$, so A2 gives search exactly when $p_2 \leq V - \frac{s}{q}$. For $v_1 = V$, the gain is $q \max\{p_1 - p_2, 0\}$, so A1 gives search exactly when $p_1 - p_2 > \frac{s}{q}$. These prove the two cutoff rules. A price $p_1 > V$ yields no sales. If $p_2 > V - \frac{s}{q}$, the gain from inspection is at most $q \max\{V - p_2, 0\} < s$, so seller 2 is never inspected. $\square$

A.2 Proof of Proposition 1

Each seller can guarantee strictly positive revenue. Seller 1 can charge any sufficiently small $0 < p_1 < \frac{s}{q}$ and sell whenever $v_1 = V$. Seller 2 can charge any $0 < p_2 < V - \frac{s}{q}$ and sell whenever $(v_1, v_2) = (0, V)$. Thus a pure equilibrium cannot involve a zero price, $p_1 > V$, or $p_2 > V - \frac{s}{q}$.

Consider the remaining domain $0 < p_1 \leq V$, $0 < p_2 \leq V - \frac{s}{q}$. If $p_1 - p_2 > \frac{s}{q}$, seller 2 can raise its price slightly while retaining demand q . If $p_1 - p_2 < \frac{s}{q}$, seller 1 can raise its price slightly while retaining demand q . At equality, seller 2 can choose a price strictly between $(1 - q)p_2$ and p_2 , increasing demand from $q(1 - q)$ to q and strictly increasing revenue. Every candidate price pair therefore admits a profitable deviation. $\square$

A.3 Proof of Theorem 1

Both proposed CDFs are increasing on their continuous parts. The first rises from zero to one. The second starts at zero and approaches $1 - \frac{s}{qV}$ at its upper bound, with the remaining mass placed at that bound. Since $0 < s < qV$, both are valid distributions with strictly positive lower bounds.

Seller 1. For $0 < p_1 \leq V$, the expected revenue against F_2 is

$$p_1 \mathbb{E}_{F_2}[D_1(p_1, p_2)] = qp_1 \left[1 - q \Pr_{F_2} \left(p_2 < p_1 - \frac{s}{q} \right) \right]. \quad (19)$$

For $\underline{p}_1 \leq p_1 < V$, substitution of F_2 gives

$$p_1 \mathbb{E}_{F_2}[D_1(p_1, p_2)] = qp_1 \left[1 - \left(1 - \frac{(1-q)V + s}{p_1} \right) \right] = q[(1-q)V + s]. \quad (20)$$

At $p_1 = V$, a good-top-match buyer stops when $p_2 = V - \frac{s}{q}$ and searches at every lower second price. Including the mass at seller 2's upper bound therefore gives

$$V \left[q \frac{s}{qV} + q(1-q) \left(1 - \frac{s}{qV} \right) \right] = q[(1-q)V + s]. \quad (21)$$

A3 preserves seller 1's sales at zero purchase surplus. For $0 < p_1 < \underline{p}_1$, good-top-match buyers never search, so revenue is $qp_1 < q[(1-q)V + s]$. At $p_1 = 0$ or $p_1 > V$, revenue is zero. Hence every price in seller 1's support is a best response against all nonnegative pure-price deviations.

Seller 2. For $0 < p_2 \leq V - \frac{s}{q}$, bad-top-match buyers search. Good-top-match buyers search if $p_1 > p_2 + \frac{s}{q}$. Thus

$$p_2 \mathbb{E}_{F_1}[D_2(p_1, p_2)] = qp_2 \left[1 - qF_1 \left(p_2 + \frac{s}{q} \right) \right]. \quad (22)$$

For $\underline{p}_2 \leq p_2 \leq \overline{p}_2$, substituting F_1 yields

$$p_2 \mathbb{E}_{F_1}[D_2(p_1, p_2)] = qp_2 \left[1 - \left(1 - \frac{(1-q)(V - \frac{s}{q})}{p_2} \right) \right] = q(1-q) \left(V - \frac{s}{q} \right). \quad (23)$$

This calculation includes the upper bound: A2 ensures that bad-top-match buyers still search there. For $0 < p_2 < \underline{p}_2$, both top-match types search at every price in seller 1's support, giving revenue $qp_2 < q(1-q)(V - \frac{s}{q})$. A zero price gives zero revenue, and a price above $V - \frac{s}{q}$ attracts no inspection. Hence every price in seller 2's support is a best response against all nonnegative pure-price deviations.

The payoff from any mixed-price deviation averages the corresponding pure-price payoffs and cannot exceed these bounds. Both optimality conditions in Definition 1 therefore hold. $\square$

A.4 Proof of Theorem 2

The two indifference calculations in (20) and (23) give the continuation revenues in (13). Their difference is

$$\begin{aligned}\mathbb{E}[\Pi_1] - \mathbb{E}[\Pi_2] &= q[(1-q)V + s] - q(1-q) \left(V - \frac{s}{q} \right) \\ &= qs + (1-q)s = s.\end{aligned}\tag{24}$$

Against an opponent bid b , winning rather than losing changes expected net profit by $s - b$. Bidding s wins when this difference is positive, loses when it is negative, and leaves the seller indifferent at equality. It is therefore weakly dominant conditional on the specified pricing continuation. With both sellers bidding s , the position payment is s and

$$\mathbb{E}[\pi_1] = q[(1-q)V + s] - s = (1-q)(qV - s) = \mathbb{E}[\pi_2].$$

The buyer is optimal after every price pair, and Theorem 1 supplies a pricing equilibrium after every auction outcome. These strategies and the symmetric bids therefore form a subgame-perfect equilibrium. $\square$

A.5 Purchase paths underlying demand

For $0 < p_1 \leq V$ and $0 < p_2 \leq V - \frac{s}{q}$, the following paths give the demand functions in (6).

(v_1, v_2)	$p_1 - p_2 > \frac{s}{q}$	$p_1 - p_2 \leq \frac{s}{q}$
$(0, 0)$	Search, then exit	Search, then exit
$(0, V)$	Search, buy 2	Search, buy 2
$(V, 0)$	Search, return to 1	Buy 1 immediately
(V, V)	Search, buy 2	Buy 1 immediately

A.6 Conditional revenues and welfare

The expectations below integrate only over independent equilibrium prices, conditional on the indicated match draws. The price means are

$$\begin{aligned}\mathbb{E}[p_1] &= \frac{s}{q} + \frac{(1-q)\left(V - \frac{s}{q}\right)}{q} \log\left(\frac{1}{1-q}\right), \\ \mathbb{E}[p_2] &= \frac{(1-q)V + s}{q} \log\left(\frac{V}{(1-q)V + s}\right).\end{aligned}\tag{25}$$

The probability of search after a good top match is

$$\Pr\left(p_1 - p_2 > \frac{s}{q}\right) = \frac{p_1 p_2}{s^2} \log\left(\frac{p_1}{(1-q)V}\right) - \frac{p_2}{s}.\tag{26}$$

When both matches are good, expected seller revenues are

$$\mathbb{E}[\Pi_1 \mid V, V] = \mathbb{E}\left[p_1 \cdot \mathbf{1}_{\{p_1 - p_2 \leq \frac{s}{q}\}}\right] = \frac{(1-q)V + s - (1-q)\mathbb{E}[p_1]}{q},\tag{27}$$

$$\mathbb{E}[\Pi_2 \mid V, V] = \mathbb{E}\left[p_2 \cdot \mathbf{1}_{\{p_1 - p_2 > \frac{s}{q}\}}\right] = \frac{(1-q)\left(V - \frac{s}{q}\right) - (1-q)\mathbb{E}[p_2]}{q}.\tag{28}$$

These identities follow by conditioning on the seller's own price and substituting the other seller's CDF. With $(v_1, v_2) = (0, V)$, only seller 2 makes a sale and earns p_2 . With $(V, 0)$, only seller 1 makes a sale and earns p_1 . With $(0, 0)$, neither makes a sale. A bad-top-match buyer always searches at equilibrium prices, whereas a good-top-match buyer searches with the probability in (26). Applying the surplus identities in (16)–(17) gives Table 5.

Table 5: Conditional equilibrium outcomes and observed match frequencies

s	(v_1, v_2)	Count	$\mathbb{E}[\Pi_1]$	$\mathbb{E}[\Pi_2]$	$\mathbb{E}[\text{CS}]$	$\mathbb{E}[\text{TS}]$
20	(0, 0)	97	0.00	0.00	-20.00	-20.00
20	(0, 100)	119	0.00	49.93	30.07	80.00
20	(100, 0)	105	81.59	0.00	13.08	94.67
20	(100, 100)	99	58.41	10.07	26.19	94.67
40	(0, 0)	112	0.00	0.00	-40.00	-40.00
40	(0, 100)	104	0.00	18.96	41.04	60.00
40	(100, 0)	113	93.86	0.00	2.91	96.77
40	(100, 100)	121	86.14	1.04	9.60	96.77

Expectations are conditional on each match pair. Counts sum to 420 and 450. Platform revenue equals s in every row, top profit is top revenue minus s , and second profit equals second revenue. Treatment benchmarks weight the rows by their observed frequencies. Values are rounded for display.

B Experimental instructions

The instructions below reproduce the low-search-cost version retained in the original manuscript. In the high-search-cost treatment, the second visit costs 40 rather than 20 points. Formatting has been adjusted for this appendix.

Welcome! Thanks for participating in this experiment on economic decision-making. Your participation is completely voluntary. Please remain seated and do not use your cellphone. You will interact anonymously through the computers. Please do not talk or look at others' screen.

At the end of the experiment, we will record your earnings amount and you will be paid by cash privately.

Fixed Role and Random Group

In this experiment, you will play the role of either a **Seller** or a **Buyer**. Your role will be assigned at the beginning and will remain fixed throughout the entire experiment.

There are **30 rounds** in total. In each round, you will be randomly grouped into a market consisting of **3 participants** (including yourself): **2 Sellers and 1 Buyer**. Groups are **randomly shuffled** every round, so you will likely interact with different people in the room from round to round. You will never know exactly who you are matched with.

Products and Fit Values

Sellers offer products to the Buyer. A seller's product can either be a Good Fit or a Bad Fit for the buyer, with equal chances.

- **Good Fit:** Value to the buyer is **100 points**. This happens with a chance of **50%**.
- **Bad Fit:** Value to the buyer is **0 points**. This happens with a chance of **50%**.

Before visiting a seller, the buyer does NOT know whether the product will be a good fit or bad fit. The fit is determined **randomly** and **independently** for each seller. The buyer only knows a product's fit value *after visiting* that seller. Knowing the fit with one seller does not provide any information about the fit with the other seller.

Decisions in Three Stages

Seller Bidding $\implies$ Seller Pricing $\implies$ Buyer Visit & Purchase

Each round proceeds according to the following three stages:

Stage I: Sellers Bid and Pay for Top Position

The two sellers simultaneously submit bids (in points) to compete for the "Top Position" on the Buyer's screen. A seller cannot see the other seller's bid when choosing their own bid. Bids must be between 0 to 100 points.

- **The Top Seller:** The seller with the higher bid wins and is displayed **first** to the buyer. It costs the buyer **0 points** to visit the **Top Seller**.
- **The Second Seller:** The seller with the lower bid loses and is displayed **second** to the buyer. It costs the buyer **20 points** to visit the **Second Seller**. The buyer decides whether

or not to pay this cost, meaning the Second Seller might not be visited at all.

Tie-breaking: If both sellers submit the exact same bid, the computer will randomly select one seller to win and become the Top Seller (each seller has an equal chance).

How the Bidding Payment Works

The **Top Seller** wins the position, but pays a position payment equal to the **Second Seller's bid** (not their own bid). The **Second Seller** pays **0 points**.

Example: Seller A bids **22**. Seller B bids **14**. Seller A becomes the Top Seller, but Seller A only pays **14 points** (not 22). Seller B becomes the Second Seller and pays **0 points**.

Stage II: Sellers Set Prices Observed by Buyer

After sellers see whether they are the Top Seller or Second Seller, they simultaneously post a price for their product. Once posted, both prices are **immediately observed** by the buyer. When setting prices, sellers do not know if their product is a Good Fit or Bad Fit for the buyer.

Stage III: Buyer Visits and Purchases

The buyer observes the prices posted by both sellers. The buyer automatically visits the **Top Seller** for free and learns its product fit.

The buyer then decides whether to pay a **Visit Cost of 20 points** to discover the fit of the **Second Seller**. The **Visit Cost is non-refundable**, even if the buyer ultimately decides not to buy from the Second Seller after the visit.

Finally, the buyer can choose to purchase one product from any seller the buyer has visited. Specifically, a buyer who visits the Second Seller can still choose to buy from the Top Seller, without paying any *additional* visit cost. The buyer may also choose to quit the market at any time and buy nothing.

Examples of Buyer's Decision Screen

For example, if sellers post prices of **X** and **Y**, the buyer observes **one** of these two screens, depending on whether Top Seller is randomly determined to have a Good Fit or Bad Fit. Notice that buyer always knows Top Seller's fit, but Second Seller's fit remains **?** until visited.

Scenario A: Top Seller is a Good Fit	Scenario B: Top Seller is a Bad Fit												
<table><tr><th>Top Seller</th><th>Second Seller</th></tr><tr><td>X points Posted Price</td><td>Y points Posted Price</td></tr><tr><td>Good Fit (100) Product Fit</td><td>? Product Fit</td></tr></table>	Top Seller	Second Seller	X points Posted Price	Y points Posted Price	Good Fit (100) Product Fit	? Product Fit	<table><tr><th>Top Seller</th><th>Second Seller</th></tr><tr><td>X points Posted Price</td><td>Y points Posted Price</td></tr><tr><td>Bad Fit (0) Product Fit</td><td>? Product Fit</td></tr></table>	Top Seller	Second Seller	X points Posted Price	Y points Posted Price	Bad Fit (0) Product Fit	? Product Fit
Top Seller	Second Seller												
X points Posted Price	Y points Posted Price												
Good Fit (100) Product Fit	? Product Fit												
Top Seller	Second Seller												
X points Posted Price	Y points Posted Price												
Bad Fit (0) Product Fit	? Product Fit												

Points and Earnings

At the end of the experiment, the computer will randomly select **ONE out of the 30 rounds**

to be your “Payment Round.” Your points in this Payment Round will be converted into dollar earnings using a conversion ratio of **10 points = \$1**.

You will start each round with an initial balance of **200 points** (worth \$20), which will be added to your payoff points in that round. This starting balance resets each round and does not carry over. Your total earnings for this experiment will be **\$10** show-up fee plus your converted points in your (randomly selected) Payment Round.

Your payoff points are calculated as follows:

Buyer Payoff Points:

- If you purchase: **Product Fit – Price Paid – Visit Cost** (if you visit Second Seller)
- If you quit without purchase: **– Visit Cost** (if you visit Second Seller)

Notice that Visit Cost is non-refundable for the buyer.

Seller Payoff Points:

- If you make a sale: **Price Charged – Position Payment**
- If you do not make a sale: **– Position Payment**

Notice that Position Payment is non-refundable for sellers.

For each round: **Total Points = 200 Points + Payoff Points**

Please be patient while waiting for others to finish their decisions. At the end of each round, you will see a feedback screen showing your payoff and total points for that round.

Below is a summary of the instructions. Next, you will answer three comprehension questions and play two practice rounds. These practice rounds help you learn the interface and do not affect your earnings. After that, we will start the formal rounds. Once you have made a choice, you cannot go back to a previous screen to change it. **If you have any questions, please raise your hand at any time during the experiment.**

Instructions Summary

Rounds	30 rounds. Your role is fixed: you are either a Seller or a Buyer .
Groups	Each round: 2 Sellers + 1 Buyer are randomly grouped.
Three Stages	Bidding $\implies$ Pricing $\implies$ Visit & Purchase
Bidding	Highest bid becomes Top Seller and pays the other Seller’s bid.
Pricing	Prices posted by both Sellers are observed by Buyer before any visit.
Product Fit	When Buyer visits a Seller, Buyer discovers if the product is a good fit worth 100 points (50% chance), or a bad fit worth 0 points (50% chance). The fit is determined independently for each seller.
Visit Cost	Visiting Top Seller is free. Visiting Second Seller costs Buyer 20 points .
\$Earnings	Show-up \$10 + Points of ONE random round (10 points = \$1).